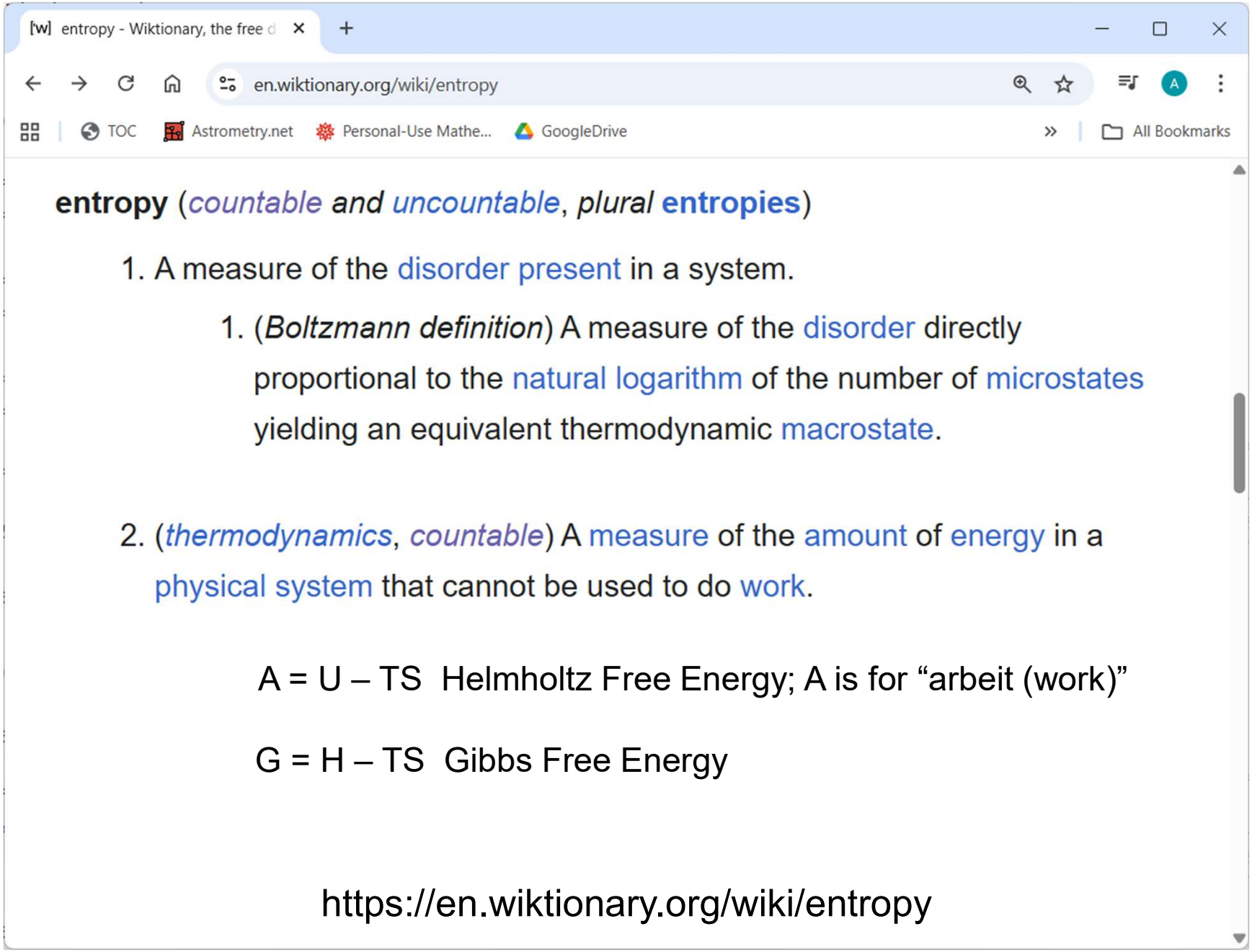
CH365 Chemical Engineering Thermodynamics

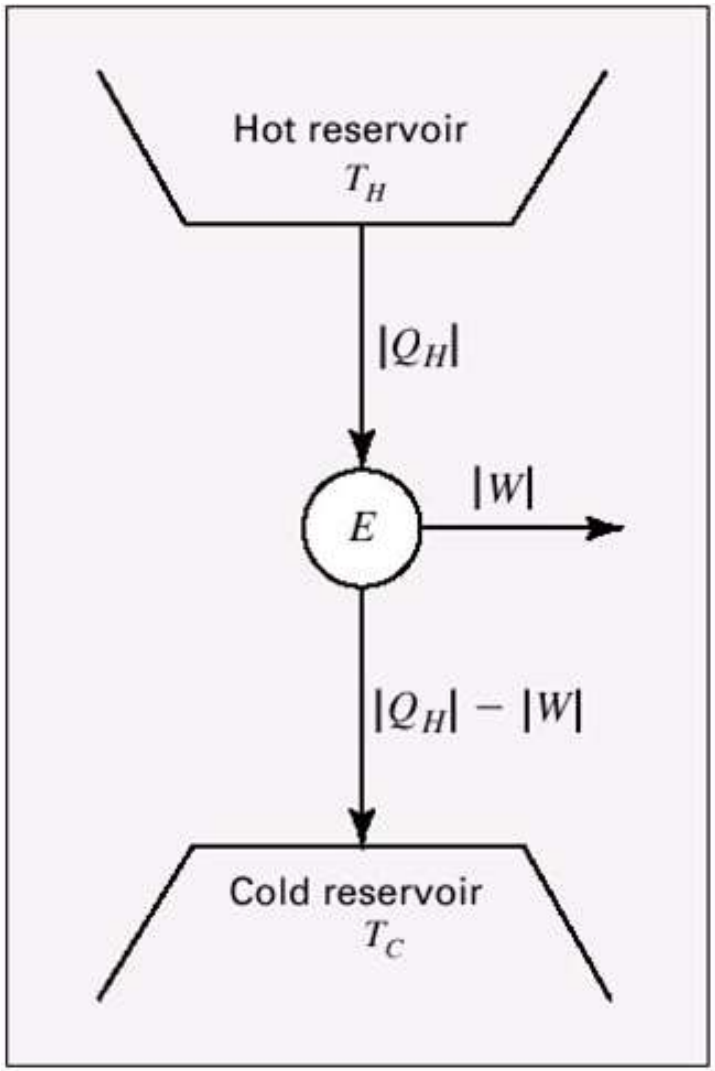
Lesson 25

3rd Law and Entropy from the Microscopic Viewpoint

What is Entropy?



Thermal Efficiency



(Eq. 5.8)

$$\eta = \frac{|W|}{|Q_H|} = 1 - \frac{T_C}{T_H}$$

What happens to η as T_C goes to zero?

As T_C goes to zero, the amount of "lost work" goes to zero.

Third Law of Thermodynamics

The absolute entropy is zero at absolute zero temperature.

This is tempting but not quite right.

The absolute entropy is zero for all perfect crystalline substances at absolute zero temperature.

Mathematical statement of the 3rd Law:

$$S = S(T) = \int_0^{T_f} \frac{(C_P)_S}{T} dT + \frac{\Delta H_f}{T_f} + \int_{T_f}^{T_v} \frac{(C_P)_L}{T} dT + \frac{\Delta H_V}{T_V} + \int_{T_V}^{T} \frac{(C_P)_G}{T} dT$$

(Eq. 5.40)

Based on Clausius axiomatic (postulate) approach (entropy is Q/T).

Allows calculation of entropy with respect to a reference temperature.

The 3rd Law repairs an important problem.

Untitled-1 * - Wolfram Mathematica 13.1

File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

Input

In[1]:=
```
(*CaO(s) m.p. 2845 K*)
(*Cp from Appendix C page 670*)
Cp = 6.104 + .443 * 10^-3 * T - 1.047 * 10^5 * T^-2;
```

In[2]:=
$$\int_0^{273} \frac{Cp}{T}\, dT$$

Integrate: Integral of $0.000443 - \frac{104700.}{T^3} + \frac{6.104}{T}$ does not converge on {0, 273}.

Out[2]=
$$\int_0^{273} \frac{6.104 - \frac{104\,700.}{T^2} + 0.000443\,T}{T}\, dT$$

In[3]:=
$$s[T_] = \int \frac{Cp}{T}\, dT$$

Out[3]=
$$\frac{52\,350.}{T^2} + 0.000443\,T + 6.104\,\text{Log}[T]$$

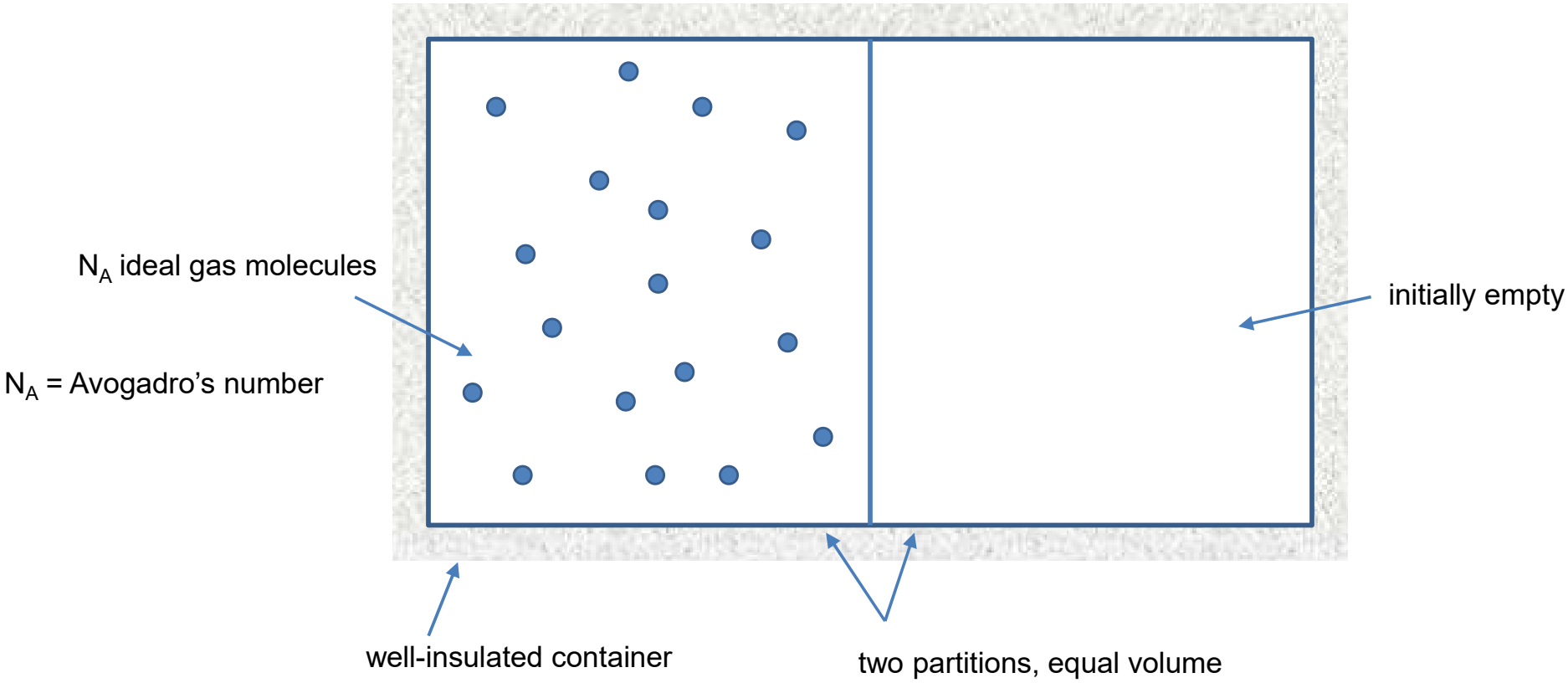
In[4]:=
```
s[273]                - s[0]
```

Out[4]= 35.0635659798

100%

Statistical Interpretation

- ideal gas
  - molecules do not interact
  - internal energy resides within the individual molecules



$$U = U(T,V)$$

ch. 4, p. 138, (un-numbered equation)

$$dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \underbrace{\left(\frac{\partial U}{\partial V}\right)_T}_{0} dV$$

0 → Because molecules don't interact

definition of C_V

$$\frac{\Delta S}{R} = \underbrace{\int_{T_{before}}^{T_{after}} \frac{C_P^{ig}}{R}\frac{dT}{T}}_{0} - \underbrace{\ln\left(\frac{P_{final}}{P_{initial}}\right)}_{}$$

0, no change in T

P drops by half

(Eq. 5.10)

ch. 3, p. 79, (Eq. 3.13a)

$$dU = C_V dT$$

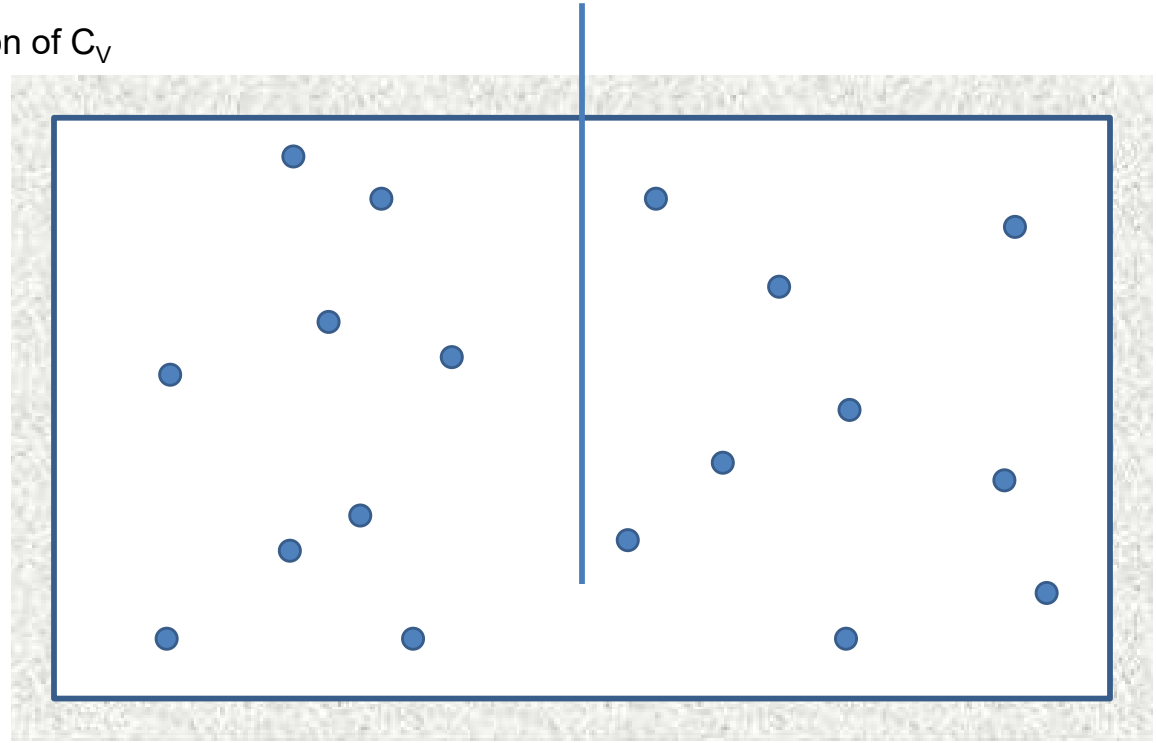
Recall:

ch. 3, p. 79, (Eq. 3.14a)

$$dH = C_P dT$$

definition of C_P:

$$C_P \equiv \left(\frac{\partial H}{\partial T}\right)_P$$



$$\Delta U = C_V \Delta T = 0$$

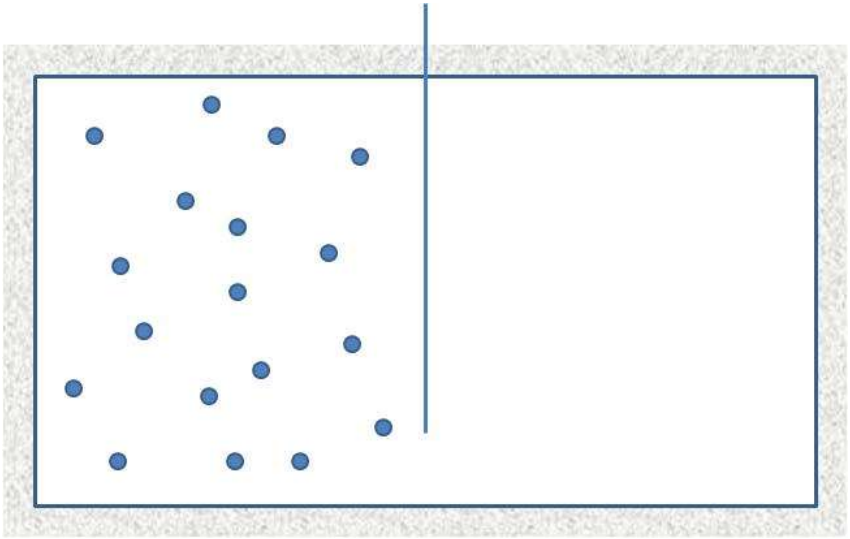
But if $\Delta U=0$, then T does not change.

$$\Delta S = -R \cdot \ln\left(\frac{P_{after}}{P_{before}}\right) = \underbrace{R \cdot \ln(2)}_{}$$

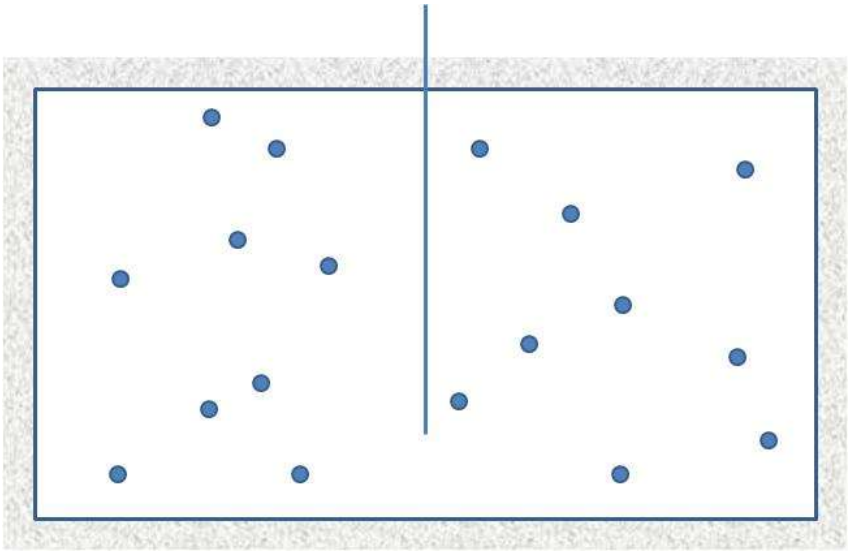
>0 (irreversible)

Result of *classical* thermodynamics

more ordered → less random → less disordered



less ordered → more random → more disordered



- immediately after opening
- molecules are not randomly distributed over the total volume
- crowded into half the space

Increasing disorder (or decreasing structure) on the molecular level corresponds to increasing entropy.

Expression for disorder postulated by J.W. Gibbs and L. Boltzmann, 1878.

Quantitative Expression of Disorder

Formulation of J.W. Gibbs and L. Boltzmann

$$\Omega = \frac{N!}{(N_1!)(N_2!)(N_3!)\cdots} \qquad \text{(Eq. 5.36)}$$

$\Omega =$ number of different ways that microscopic particles can be distributed among the "states" accesible to them.

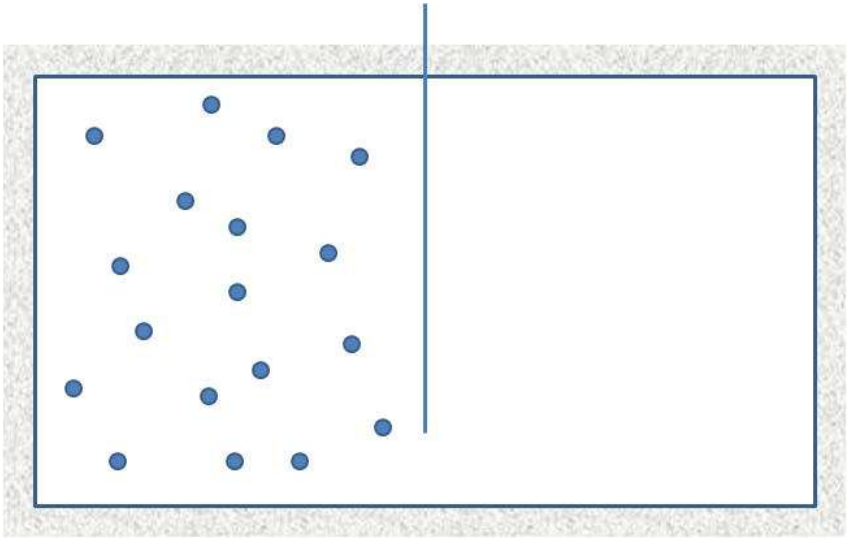
$N =$ total number of particles

$N_1 =$ number of particles in state 1

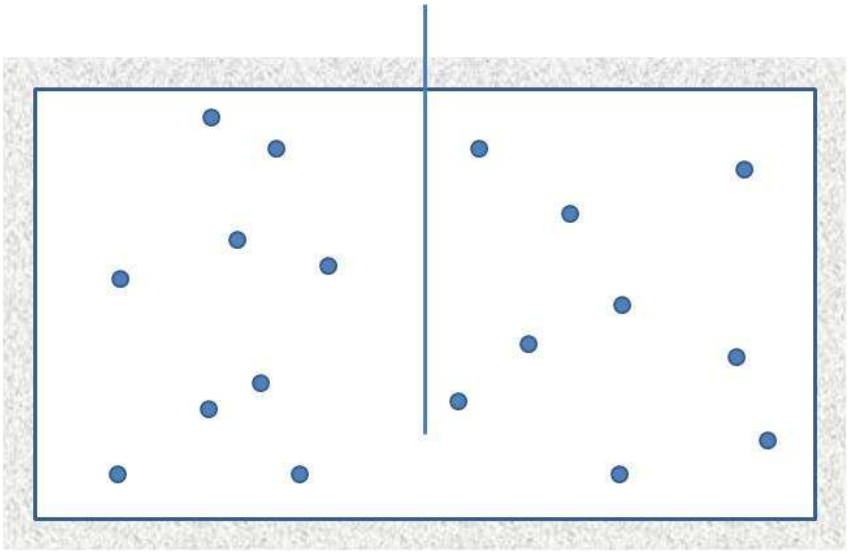
$N_i =$ number of particles in state i

Boltzmann used “W” for “Wahrscheinlichkeit”

more ordered = less random = less disordered



less ordered = more random = more disordered



All molecules are in one of the two states.

$$\Omega_{\text{initial}} = \frac{N_A!}{(N_A!)(0!)}$$

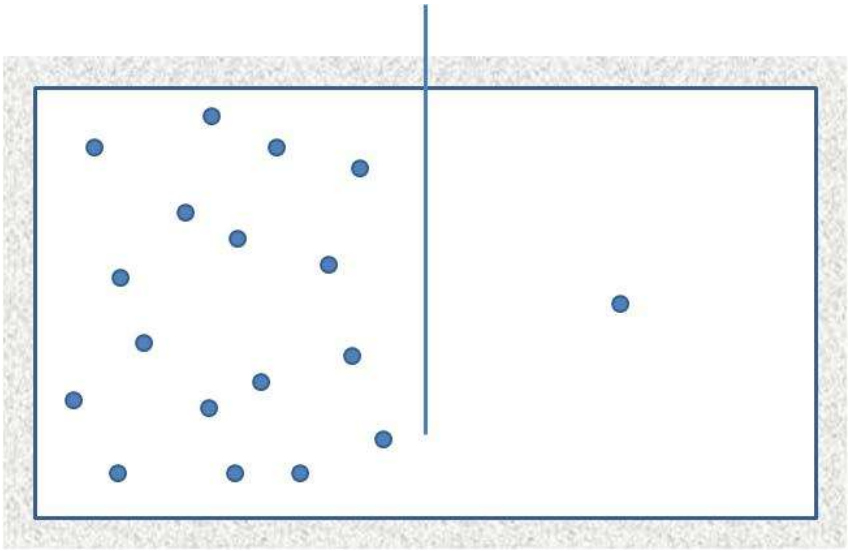
$$= \frac{18!}{(18!)(0!)}$$

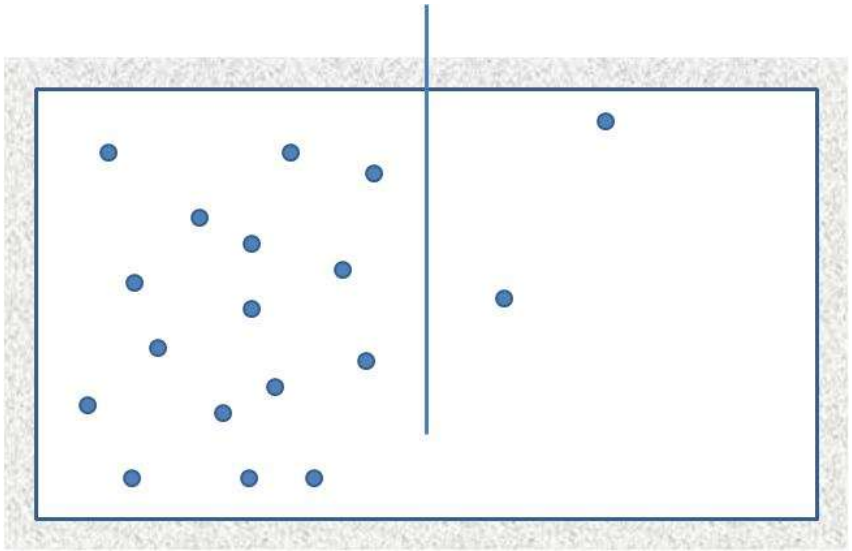
$$= 1$$

$$\Omega_{\text{final}} = \frac{N_A!}{\left(\frac{N_A}{2}!\right)\left(\frac{N_A}{2}!\right)}$$

$$= \frac{18!}{9! \cdot 9!}$$

$$= 48,620$$





$$\Omega_1 = \frac{18!}{(17!)(1!)} = 18$$

$$\Omega_2 = \frac{18!}{(16!)(2!)} = 153$$

unbounded as N_A increases

for the 18 particles, $\Omega_{final} = 48,620$

How about $N_A = 10^{23}$?

There are a large number of ways that particles can distribute themselves *uniformly* between the two states.

There are also a large number of ways that particles can distribute themselves *non-uniformly* between the two states.

The ratio of a specific Ω_i to the total Ω is the probability of that distribution.

Boltzmann's connection:

$$S_i = k \cdot \ln(\Omega_i)$$

(Eq. 5.37)

Equations 5.36 and 5.37 are the basis of modern statistical thermodynamics

Boltzmann's constant

Boltzmann had what Jules Winnfield called "a moment of clarity." He realized that equation 5.37 unified the statistical and classical approaches to entropy.

Boltzmann's thought process:

$$S_i = k \cdot \ln(\Omega_i) \quad \Rightarrow \quad S_{final} - S_{initial} = k \cdot \ln\left(\frac{\Omega_{final}}{\Omega_{initial}}\right)$$

(Eq. 5.37)

$$\Omega_{final} = \frac{N_A!}{\left(\frac{N_A}{2}!\right)^2} \qquad \Omega_{initial} = 1$$

$$S_{final} - S_{initial} = k \cdot \ln\left(\frac{N_A!}{\left(\frac{N_A}{2}!\right)^2}\right) = k\left[\ln(N_A!) - \ln\left(\frac{N_A}{2}!\right)^2\right] = k\left[\ln(N_A!) - 2 \cdot \ln\left(\frac{N_A}{2}!\right)\right]$$

James Stirling
(1692-1770)

Stirling's Formula: $\ln(x!) = x \cdot \ln(x) - x$ Limit of large x

$$S_{final} - S_{initial} = k\left[\left(N_A \cdot \ln(N_A) - N_A\right) - 2 \cdot \left(\frac{N_A}{2}\ln\left(\frac{N_A}{2}\right) - \frac{N_A}{2}\right)\right]$$

Boltzmann's thought process, continued:

$$S_{final} - S_{initial} = k\left[\left(N_A \cdot \ln\left(N_A\right) - N_A\right) - 2 \cdot \left(\frac{N_A}{2}\ln\left(\frac{N_A}{2}\right) - \frac{N_A}{2}\right)\right]$$

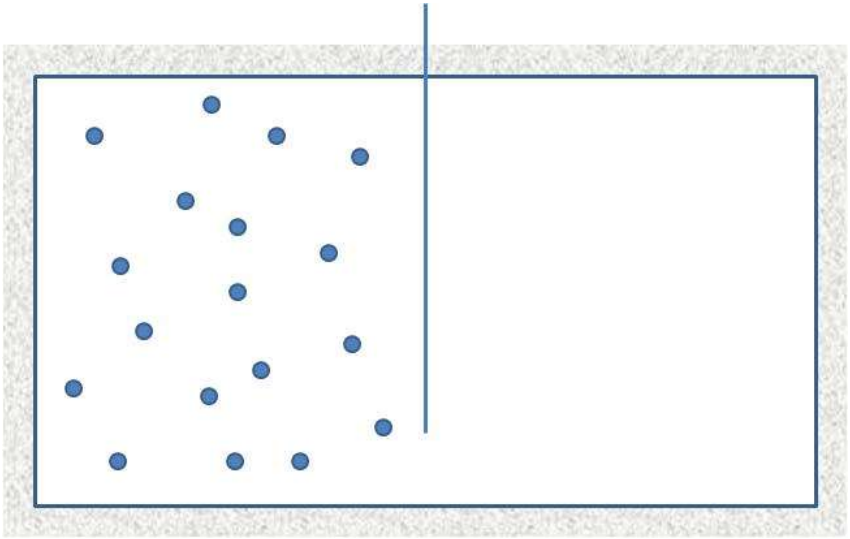
$$= k\left[N_A \cdot \ln\left(N_A\right) - N_A - N_A \cdot \ln\left(\frac{N_A}{2}\right) + N_A\right]$$

$$= k\left[N_A \cdot \ln\left(N_A\right) - N_A \cdot \ln\left(\frac{N_A}{2}\right)\right]$$

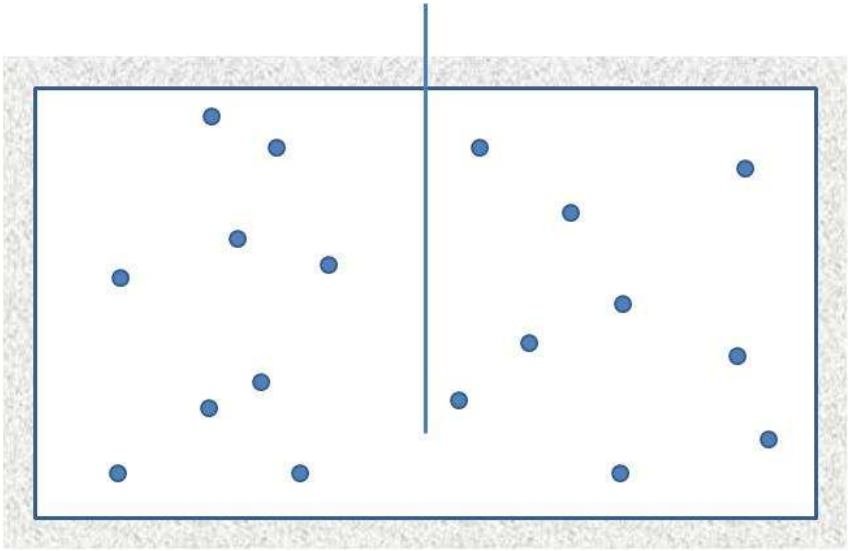
$$= k \cdot N_A \cdot \left[\ln\left(N_A\right) - \ln\left(\frac{N_A}{2}\right)\right]$$

$$= k \cdot N_A \cdot \ln\left(\frac{N_A}{\frac{N_A}{2}}\right) = k \cdot N_A \cdot \ln(2)$$

more ordered = less random = less disordered



less ordered = more random = more disordered



- immediately after opening
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Increasing disorder (or decreasing structure) on the molecular level corresponds to increasing entropy.



https://en.wikipedia.org/wiki/File:Zentralfriedhof_Vienna_-_Boltzmann.JPG

Questions?